

SemesterMate

A student productivity and academic management mobile app - Mobile Application Development, 6th Semester

The problem

University students juggle CT exams, lab exams, vivas, assignments, finals, attendance, semester fees and GPA across notebooks, sticky notes and memory. Deadlines slip; attendance drops below 75% silently; GPA is recalculated by hand each semester. There is no single place for all of it.

The solution

SemesterMate is a single Android app that centralises every academic obligation. Data is synced to the cloud (survives phone changes), protected by per-user Row Level Security at the database layer, and augmented with on-device reminders so deadlines are never missed.

Key features

- Authentication — email/password via Supabase Auth, persistent sessions, password reset.
- Dashboard — greeting, upcoming exams, fee due, attendance % and CGPA at a glance, weekly productivity chart.
- Task system — full CRUD over CT / Lab / Viva / Assignment / Final, with priorities, status, search, type filters, list and calendar views.
- Attendance tracker — per-subject classes, auto-percentage, <75% warning, analytics bar chart with a 75% guideline.
- Semester fee tracker — total / paid / due, deadline reminders, progress bars, overdue flags.
- GPA and CGPA — add courses with credits and grades, auto semester GPA, cumulative CGPA, multi-semester trend line.
- Local notifications — one-hour-before reminders for tasks, one-day-before reminders for fee deadlines.
- Profile and settings — editable profile, light/dark theme, notification toggle.

Tech stack

Frontend	Flutter 3.32 (Dart 3), Material 3, Google Fonts (Inter)
State	Riverpod (AsyncNotifier + Provider)
Routing	go_router with auth guard and ShellRoute
Backend	Supabase: PostgreSQL + GoTrue Auth
Security	Row Level Security policies on every table
Charts	fl_chart (bar and line)
Calendar	table_calendar
Notifications	flutter_local_notifications + timezone
DevOps	GitHub Actions: release APK built and uploaded per push

Architecture

Clean three-layer architecture with strict separation of concerns:

- Models — pure Dart data classes with fromMap / toInsertMap for Supabase JSON.
- Repositories — single source of truth for each entity; wrap the Supabase client.
- Providers (Riverpod) — reactive state; AsyncNotifier exposes loading / data / error.
- Screens and widgets — UI consumes providers; never talks to the DB directly.
- Core — theme, env (via --dart-define), constants, validators.

Database and security

PostgreSQL schema with five tables (profiles, tasks, attendance, semester_fees, gpa_records) plus a trigger that auto-creates a profile row on signup. Every table has Row Level Security enabled with policies of the form:

```
create policy "tasks_select_own" on tasks for select using (auth.uid() = user_id);
```

Even with the anon key embedded in the APK, a malicious user cannot read another user's data — Postgres enforces ownership at the row level. The privileged service_role key never ships with the app.

Build and deployment

- GitHub Actions regenerates the android/ folder on every build, patches the manifest for notification permissions, enables core library desugaring, and runs flutter build apk --release.
- Secrets (SUPABASE_URL, SUPABASE_ANON_KEY) are injected via --dart-define from GitHub repository secrets — never committed to source.
- The release APK is uploaded as a GitHub artifact, downloadable from the Actions UI.

For a JavaScript developer's mind

JSX component tree	Flutter widget tree (everything is a Widget)
React Context + reducers	Riverpod providers + AsyncNotifier
React Router	go_router (declarative, with redirect guards)
Tailwind / shadcn	Material 3 ThemeData + custom theme.dart
npm + package.json	pub.dev + pubspec.yaml
recharts / chart.js	fl_chart
Vite hot reload	Flutter hot reload (sub-second)
Supabase JS SDK	supabase_flutter (same primitives, Dart bindings)

Outcome

A production-style, demo-ready Android app: signed release APK built from a 64-file Flutter codebase (~6,700 lines of Dart), end-to-end Supabase integration with Row Level Security, six full feature modules, light/dark theme, local notifications, calendar and chart visualisations — all buildable in under four minutes on GitHub-hosted CI.

Source: github.com/Akram-hossain/Smart-Queue-Mobile-App APK: Actions tab > latest run > Artifacts > semestermate-release-apk